

Anas MADANI

Full Stack Software Engineer

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PROFILE

Full-stack software engineer (ENSIAS) specialized in microservices architectures and DevOps practices (Docker, Kafka, CI/CD). Hands-on experience in banking and industrial environments, from design through infrastructure industrialization, with a strong focus on code quality.

EDUCATION

- **École Nationale Supérieure d'Informatique et d'Analyse des Systèmes (ENSIAS)** Rabat, Morocco
Engineering Degree – Software Engineering 2022 – 2025
- **CPGE – Settati** Settati, Morocco
Preparatory Classes for Engineering Schools, TSI 2019 – 2022

PROFESSIONAL EXPERIENCE

- **Banque Centrale Populaire (BCP)** Casablanca, Morocco
Software Engineer Intern (Graduation Project) February 2025 – July 2025
 - Designed and developed a microservices architecture for a banking recommendation platform, with a Java/Spring Boot backend and a React/TypeScript frontend.
 - Industrialized the infrastructure (Docker, Kafka, Redis, Keycloak) within a team in an Agile/Scrum environment.
 - *Stack:* Java, Spring Boot/Cloud, React, TypeScript, Python, FastAPI, PostgreSQL, Redis, Kafka, Keycloak, Docker, Git
- **N+ONE DATACENTERS** Casablanca, Morocco
Software Engineer Intern August 2024 – September 2024
 - Developed a full-stack platform (Django / React) containerized with Docker Compose (backend, frontend, Nginx reverse proxy, PostgreSQL).
 - Set up a CI/CD pipeline (GitHub Actions) and an asynchronous monitoring module (scheduler).
 - Implemented cryptographic document security via AES-256-GCM encryption.
 - *Stack:* Python, Django REST Framework, React, Vite, PostgreSQL, Docker, Nginx, GitHub Actions, AES-GCM
- **Commune Er-rich** Er-rich, Morocco
Software Engineer Intern July 2023 – August 2023
 - Independently implemented a worker management application to digitalize the municipality's administrative processes.
 - Led the transition from paper-based systems to a centralized digital solution.
 - *Stack:* Java, Spring Boot, React, Tailwind CSS, PostgreSQL, Git

PROJECTS

- **BusConnect:** Bus reservation platform built on a microservices architecture with Spring Boot 3, Spring Cloud (Eureka, API Gateway), React 18, and Kubernetes deployment. Integrates PostgreSQL, MySQL, and MongoDB depending on the service. [GitHub](#)
- **MLOps Pipeline Generator:** MLOps code generator based on Model-Driven Engineering (MDE) using Eclipse Ecore and Acceleo. Automatically generates Python scripts, Dockerfiles, and Kubernetes configurations from metamodels via an SWT/EMF Forms interface. [GitHub](#)
- **BacSurveillance:** Exam supervision management platform using a greedy + backtracking algorithm for automatic teacher assignment. FastAPI/PostgreSQL backend, React 19 (TypeScript) frontend, Word/Excel export, and bilingual Arabic/French interface. [GitHub](#)
- **FitHub:** Full-stack gym management application built on a microservices architecture with CI/CD pipelines via GitHub Actions. [GitHub](#)

TECHNICAL SKILLS

- **Backend:** Spring Boot, Spring Cloud, FastAPI, Django REST Framework
- Frontend:** React, TypeScript, Tailwind CSS, Vite
- DevOps & Infra:** Docker, Kafka, GitHub Actions, Nginx, Linux, Kubernetes
- Databases:** PostgreSQL, MongoDB, MySQL, Redis
- Languages (Programming):** C, C++, Python, Java, JavaScript, SQL
- Spoken Languages:** Arabic (native), French (fluent), English (fluent)